

- 4 (a)** clamped horizontal wire over pulley or vertical wire attached to ceiling with mass attached (B1)  
 details: reference mark on wire with fixed scale alongside (B1) [2]
- (b)** measure original length of wire to reference mark with metre ruler / tape (B1)  
 measure diameter with micrometer / digital calipers (B1)  
 measure initial and final reading (for extension) with metre ruler or other suitable scale (B1)  
 measure / record mass or weight used for the extension (B1)  
 good physics method:  
 measure diameter in several places / remove load and check wire returns to original length / take several readings with different loads (B1)
- MAX of 4 points (B4) [4]
- (c)** determine extension from final and initial readings (B1)  
 plot a graph of force against extension (B1)  
 determine gradient of graph for  $F / e$  (B1)  
 calculate area from  $\pi d^2 / 4$  (B1)  
 calculate  $E$  from  $E = F l / e A$  or gradient  $\times l / A$  (B1)
- MAX of 4 points (B4) [4]

|               |                                       |                 |              |
|---------------|---------------------------------------|-----------------|--------------|
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**5** (a) (i) energy converted from chemical to electrical when charge flows through cell